

MATH 231 – Linear Algebra I

The Vector Calculus Unit, Part 2 ■ n Dimensions, the Jacobian, and the Hessian

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About this document. Part 1 of this unit developed the whole of differential calculus for functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$: the two partial derivatives, the gradient, the directional derivative, steepest ascent, and the linear approximation. It stayed in the plane so that every statement could be drawn.

This part does two things. Sections §10–§13 lift all of that to $\mathbb{R}^n$, and they are short on purpose — the point is that almost nothing changes. Every definition is a Part 1 definition with n in place of 2; every argument is a Part 1 argument copied out. The single new ingredient is a piece of notation that lets n separate limits be written as one.

Sections §14–§15 are new. Differentiating $f : \mathbb{R}^n \rightarrow \mathbb{R}$ produced the gradient, which is *not* a function of the same type as f — so differentiating a second time needs a derivative for functions with vector output. That is the Jacobian, and applying it to the gradient produces the Hessian: the second derivative of a scalar function of n variables, which turns out to be a matrix.

A common confusion (Section numbers). This part continues the numbering of Part 1, so it opens at §10. A bare “§12” below means §12 *of this unit*, not of the vector unit, which also has a §12. Whenever another unit is meant, it is named.

Assumed. All of Part 1. From the vector unit: $\mathbb{R}^n$ and its vectors (§24.1); the standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ (§16, §24.1); the dot product and the identity $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$ (§11.3); and the Euclidean norm (§2). From the matrix unit: matrices, the transpose, and what it means for a matrix to be symmetric (§6.5).

What you should be able to do afterwards. Read the signature $f : \mathbb{R}^n \rightarrow \mathbb{R}$; write the partial derivative as a single limit using $\mathbf{e}_j$; assemble the n partials into a gradient and compute one; state the directional derivative and the steepest-ascent property in $\mathbb{R}^n$ and say why the Part 1 argument needed no change; write the linear approximation in n variables; say what a function $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is; build its Jacobian and read its shape; explain why the gradient has a Jacobian of its own; build the Hessian of a scalar function; state Clairaut’s theorem and the role of its hypothesis; and say why the Hessian is symmetric when a general Jacobian is not.

10. From the plane to $\mathbb{R}^n$

10.1. The signature

Definition 10.1 (Scalar function of n variables). A **scalar function of n variables** is a function

$$f : \mathbb{R}^n \longrightarrow \mathbb{R},$$

taking a vector $\mathbf{x} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n$ and returning a single real number $f(\mathbf{x})$.

This is §5.1 with n in place of 2, and nothing else is different: the input is a vector, the output is a scalar, and such functions are still called **scalar fields**. What is lost is the picture. A function $\mathbb{R}^2 \rightarrow \mathbb{R}$ has a graph in three dimensions — a surface — and every statement of Part 1 could be checked against it. For $n = 3$ the graph would need four dimensions, and there is no drawing it.

That loss is worth naming, because it is the reason this part is organised the way it is. The formulas survive; the pictures do not. When you want to know what a statement about $\mathbb{R}^n$ *means*, set $n = 2$, draw it, and trust that the algebra says the same thing with more coordinates.

10.2. Examples

Example 10.1. Each of the following is a function $\mathbb{R}^n \rightarrow \mathbb{R}$.

(a) $f(x, y, z) = x^2 + y^2 + z^2$, with $n = 3$. Three inputs, still one output.

(b) $f(\mathbf{x}) = \|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}$, for any n . The Euclidean norm eats a vector and returns its length, a single number (vector unit §2, §24.1).

(c) $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle = a_1x_1 + \cdots + a_nx_n$ for a fixed vector $\mathbf{a} \in \mathbb{R}^n$. The dot product against a fixed vector is likewise $\mathbb{R}^n \rightarrow \mathbb{R}$ (vector unit §11.3).

(d) The temperature at each point of a room: $n = 3$. The closing price of each of 500 stocks fed into a single portfolio value: $n = 500$, and no picture at all.

Remark. Examples (b) and (c) were already on the list in §5.2 for $n = 2$. They are listed again because they are the two functions $\mathbb{R}^n \rightarrow \mathbb{R}$ you will meet most often, and because their gradients, computed in §12.2, are worth knowing by heart.

10.3. Stepping in one coordinate direction

Part 1 defined the two partial derivatives with two separate limits, one displacing x and one displacing y :

$$\lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}, \quad \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}.$$

Writing n of those out, one per coordinate, is unworkable past about three. The vector unit supplies exactly the notation needed to collapse them into one.

From the vector unit. $\mathbf{e}_j$ is the j -th standard basis vector of $\mathbb{R}^n$: a 1 in slot j and 0 everywhere else (vector unit §16, §24.1). So

$$\mathbf{x} + h\mathbf{e}_j = [x_1, \dots, x_{j-1}, x_j + h, x_{j+1}, \dots, x_n]:$$

adding $h\mathbf{e}_j$ changes the j -th coordinate by h and leaves every other coordinate exactly where it was.

That is the whole of the new notation. In $\mathbb{R}^2$, $\mathbf{e}_1 = [1, 0]$ and $\mathbf{e}_2 = [0, 1]$, so $\mathbf{x} + h\mathbf{e}_1 = (x+h, y)$ and $\mathbf{x} + h\mathbf{e}_2 = (x, y+h)$ — the two displacements Part 1 wrote out by hand.

11. Partial derivatives in n variables

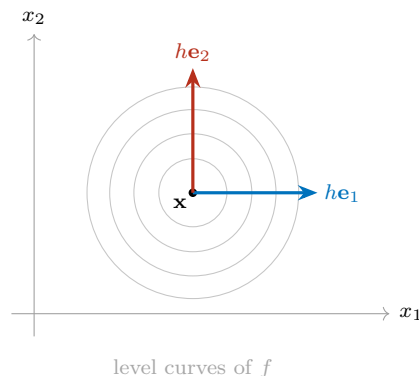
11.1. The definition

Definition 11.1 (Partial derivative). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The **partial derivative of f with respect to x_j** at the point $\mathbf{x}$ is

$$\frac{\partial f}{\partial x_j}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_j) - f(\mathbf{x})}{h}, \quad (1)$$

provided the limit exists.

One formula now covers all n of them. Setting $n = 2$ and $j = 1$ recovers the first of the two limits displayed in §10.3, and $j = 2$ recovers the second; no content was added in passing from two variables to n , only notation.



The figure can only be drawn for $n = 2$, where there are two arrows. For general n there are n of them, mutually orthogonal, and the picture is gone — but (1) is unchanged.

Remark (What the geometric reading becomes). Part 1's §6.2 read $\partial f/\partial x$ as the slope of a slice: hold $y = b$, cut the surface with that vertical plane, and take the ordinary tangent slope of the resulting curve. The same reading survives verbatim. Fixing every coordinate but x_j turns f into a one-variable function

$$g(t) = f(x_1, \dots, x_{j-1}, t, x_{j+1}, \dots, x_n),$$

and $\partial f/\partial x_j(\mathbf{x})$ is $g'(x_j)$. What is lost is only the ability to draw the slice, not the fact that it is one.

11.2. Computing them

The rule of §6.3 is unchanged, and for the same reason: every variable except x_j is held fixed, so they behave as constants and the expression becomes a function of the one live variable.

To compute $\partial f/\partial x_j$: treat every variable other than x_j as a constant, and differentiate as in one-variable calculus.

Example 11.1. Let $f(x, y, z) = x^2y + y^2z + z^2x$. Three variables, so three partials. For each, two of the three are constants:

$$\frac{\partial f}{\partial x} = 2xy + z^2, \quad \frac{\partial f}{\partial y} = x^2 + 2yz, \quad \frac{\partial f}{\partial z} = y^2 + 2zx.$$

Example 11.2. Let $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle = a_1x_1 + \dots + a_nx_n$, example (c) of §10.2. Only the j -th term involves x_j , and in it x_j appears to the first power with coefficient a_j ; every other term is a constant. So

$$\frac{\partial f}{\partial x_j} = a_j \quad \text{for every } j.$$

The partial derivatives of a dot product against $\mathbf{a}$ are the components of $\mathbf{a}$ itself — the same answer §6.3 got for $n = 2$, arrived at the same way.

Example 11.3. Let $f(\mathbf{x}) = \|\mathbf{x}\| = (x_1^2 + \dots + x_n^2)^{1/2}$, example (b) of §10.2, at any $\mathbf{x} \neq \mathbf{0}$. Holding every coordinate but x_j fixed and applying the chain rule to the outer square root,

$$\frac{\partial f}{\partial x_j} = \frac{1}{2}(x_1^2 + \dots + x_n^2)^{-1/2} \cdot 2x_j = \frac{x_j}{\|\mathbf{x}\|}.$$

Only the j -th term of the sum contributed to the inner derivative; the rest were constants.

11.3. What kind of object is a partial derivative?

Exactly as in §6.4. Fix j . At each point $\mathbf{x} \in \mathbb{R}^n$ the limit (1) produces a single number, so $\partial f / \partial x_j$ is itself a function

$$\frac{\partial f}{\partial x_j} : \mathbb{R}^n \longrightarrow \mathbb{R},$$

of the same type as f . Differentiating with respect to one variable does not reduce the number of variables.

And there are n of them, one for each j from 1 to n .

12. The gradient in $\mathbb{R}^n$

12.1. Collecting the partials

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has n derivatives, not one, and carrying n separate functions around and naming them individually would be unmanageable for any interesting n . Package them, as §7.1 packaged two.

Definition 12.1 (Gradient). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ have all n partial derivatives at $\mathbf{x}$. The **gradient** of f at $\mathbf{x}$ is the vector in $\mathbb{R}^n$ whose j -th component is the j -th partial derivative:

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}(\mathbf{x}), \frac{\partial f}{\partial x_2}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right].$$

Note the signature, and keep it in view — §14.3 comes back to it:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \text{but} \quad \nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n. \quad (2)$$

The gradient takes a point and returns a *vector*. This is the genuine departure from the one-variable case, where f' had the same signature as f , and it is what makes the second derivative in §15 a matrix rather than a number.

12.2. Examples

Example 12.1. For $f(x, y, z) = x^2y + y^2z + z^2x$, the partials were computed in §11.2, so

$$\nabla f(x, y, z) = [2xy + z^2, \quad x^2 + 2yz, \quad y^2 + 2zx],$$

a vector in $\mathbb{R}^3$. At $(1, 1, 1)$ it is $[3, 3, 3]$.

Example 12.2. For $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle$, §11.2 gave $\partial f / \partial x_j = a_j$ for every j . Assembling them,

$$\nabla f(\mathbf{x}) = [a_1, a_2, \dots, a_n] = \mathbf{a}.$$

The gradient is $\mathbf{a}$ itself, the same at every point — the n -variable counterpart of the fact that a line of slope m has derivative m everywhere.

Example 12.3. For $f(\mathbf{x}) = \|\mathbf{x}\|$, §11.2 gave $\partial f / \partial x_j = x_j / \|\mathbf{x}\|$, so for $\mathbf{x} \neq \mathbf{0}$,

$$\nabla f(\mathbf{x}) = \frac{1}{\|\mathbf{x}\|} [x_1, \dots, x_n] = \frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

The gradient of the norm is the normalisation of $\mathbf{x}$ — a unit vector pointing straight away from the origin. Read through §12.3, that says the fastest way to increase your distance from the origin is to walk directly away from it, at a rate of one unit of distance per unit of distance walked. Both halves of that sentence are what the formula says, and both are obvious, which is a good sign.

Remark. The last example is the first gradient in this unit that could not have been computed in Part 1 in any useful generality, and it is worth noticing how little work it took. The packaging is doing the labour: n partials, each a fraction, collapse into one recognisable vector.

12.3. The directional derivative and steepest ascent

Everything in §8 carries over to $\mathbb{R}^n$ with no change at all, and it is worth being precise about why.

For a unit vector $\mathbf{u} \in \mathbb{R}^n$, the **directional derivative** of f at $\mathbf{x}$ in the direction $\mathbf{u}$ is defined by the same limit as in §8.1,

$$D_{\mathbf{u}}f(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{u}) - f(\mathbf{x})}{h},$$

and it is computed from the gradient exactly as in §8.2:

$$D_{\mathbf{u}}f(\mathbf{x}) = \langle \nabla f(\mathbf{x}), \mathbf{u} \rangle = \frac{\partial f}{\partial x_1}(\mathbf{x}) u_1 + \cdots + \frac{\partial f}{\partial x_n}(\mathbf{x}) u_n.$$

Taking $\mathbf{u} = \mathbf{e}_j$ recovers the j -th partial derivative, which is the n -dimensional version of the observation that $[1, 0]$ and $[0, 1]$ recovered the two partials in Part 1.

Now run the argument of §8.3 again, and watch how much of it mentions the number of coordinates. Write the dot product in terms of the angle θ between $\nabla f(\mathbf{x})$ and $\mathbf{u}$:

$$D_{\mathbf{u}}f(\mathbf{x}) = \|\nabla f(\mathbf{x})\| \|\mathbf{u}\| \cos \theta = \|\nabla f(\mathbf{x})\| \cos \theta.$$

As $\mathbf{u}$ ranges over the unit vectors, the factor $\|\nabla f(\mathbf{x})\|$ is fixed and only $\cos \theta$ varies; it is largest, equal to 1, when $\theta = 0$; and the unique unit vector achieving that is $\mathbf{u} = \nabla f(\mathbf{x}) / \|\nabla f(\mathbf{x})\|$, which returns the value $\|\nabla f(\mathbf{x})\|$.

Not one step of that used $n = 2$. The identity $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$ holds in $\mathbb{R}^n$, $\langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{w}\|^2$ holds in $\mathbb{R}^n$, and $\cos \theta$ is still largest at $\theta = 0$. So the conclusion holds in $\mathbb{R}^n$ too.

At a point $\mathbf{x} \in \mathbb{R}^n$ where all n partial derivatives exist and $\nabla f(\mathbf{x}) \neq \mathbf{0}$, the gradient points in the direction of **steepest ascent** of f and $\|\nabla f(\mathbf{x})\|$ is the rate of climb; $-\nabla f(\mathbf{x})$ points in the direction of **steepest descent**; and the rate of change is zero exactly along the directions orthogonal to the gradient.

Remark (Why this matters more in $\mathbb{R}^n$ than in $\mathbb{R}^2$). In the plane, gradient descent competes with simply looking at the surface. In $\mathbb{R}^n$ for large n there is nothing to look at, and no way to search the directions one at a time: the unit vectors of $\mathbb{R}^{500}$ cannot be enumerated. What the boxed statement provides is a way to find the best of infinitely many directions by computing n numbers. That is the entire reason the gradient is the object modern optimisation is built on, and the reason the packaging move of §12.1 was worth making.

13. Linear approximation in n variables

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and let $\mathbf{a} \in \mathbb{R}^n$ be the point we are near. Every piece of the two-variable formula of §9.2 survives; only the number of coordinates changes.

Definition 13.1 (Linear approximation in n variables). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ have a gradient at $\mathbf{a}$. The **linear approximation** of f at $\mathbf{a}$ is

$$L(\mathbf{x}) = f(\mathbf{a}) + \langle \nabla f(\mathbf{a}), \mathbf{x} - \mathbf{a} \rangle, \tag{3}$$

and $f(\mathbf{x}) \approx L(\mathbf{x})$ for $\mathbf{x}$ near $\mathbf{a}$.

Set the three cases side by side:

$$\underbrace{f(a) + f'(a)(x - a)}_{n=1} \longleftrightarrow \underbrace{f(\mathbf{a}) + \langle \nabla f(\mathbf{a}), \mathbf{x} - \mathbf{a} \rangle}_{n=2} \longleftrightarrow \underbrace{f(\mathbf{a}) + \langle \nabla f(\mathbf{a}), \mathbf{x} - \mathbf{a} \rangle}_{\text{general } n}.$$

The last two are the same symbols. Only the length of the vectors inside the dot product has changed, and the formula never mentioned it. When $n = 1$ the dot product of two one-component vectors *is* ordinary multiplication, so (3) contains all three.

Written out in coordinates, (3) reads

$$L(\mathbf{x}) = f(\mathbf{a}) + \frac{\partial f}{\partial x_1}(\mathbf{a})(x_1 - a_1) + \cdots + \frac{\partial f}{\partial x_n}(\mathbf{a})(x_n - a_n),$$

one term per direction: each partial derivative converts the displacement in its own coordinate into a contribution to the change in f , and the contributions are added up.

Example 13.1. Let $f(x, y, z) = x^2y + y^2z + z^2x$ and approximate near $\mathbf{a} = (1, 1, 1)$. There $f(\mathbf{a}) = 3$, and §12.2 gave $\nabla f(1, 1, 1) = [3, 3, 3]$, so

$$L(x, y, z) = 3 + 3(x - 1) + 3(y - 1) + 3(z - 1).$$

At $(1.1, 0.9, 1.05)$ this predicts $3 + 3(0.1) + 3(-0.1) + 3(0.05) = 3.15$, against the true value 3.15225 — and the three displacements partly cancel, exactly as the three terms show.

Remark (What the graph of L is). For $n = 1$ the linear approximation is a tangent *line*; for $n = 2$, a tangent *plane*. For general n it is a tangent **hyperplane** — the flat object of dimension n that best matches f at $\mathbf{a}$. It cannot be drawn past $n = 2$, and it does not need to be: (3) is what it is.

Remark (Why the gradient deserves to be called the derivative). (3) is the strongest answer to the question of what replaces f' in several variables. In one variable, $f'(a)$ is the number that makes the best linear approximation work. In n variables, $\nabla f(\mathbf{a})$ is the vector that plays that role — the same job, the same position in the formula, a different type of object. Collecting the partials into a vector was not bookkeeping after all: it produced the thing that behaves like a derivative.

Looking ahead. (3) is also where the next ideas come from. Asking where L is flat — where $\nabla f(\mathbf{a}) = \mathbf{0}$ — locates the candidates for maxima and minima, which are the points at which gradient descent stalls. And approximating with something better than a flat object requires a *second* derivative, which by (2) means differentiating a function whose output is a vector. The next section defines that; the one after applies it to the gradient.

14. The Jacobian

Differentiating $f : \mathbb{R}^n \rightarrow \mathbb{R}$ produced ∇f , which by (2) is not a function of the same type as f : it takes a point of $\mathbb{R}^n$ and returns a *vector* in $\mathbb{R}^n$. To differentiate a second time we first need to know what the derivative of such a function even is.

14.1. Functions with vector output

So far the output has always been a single number. Allow it to be a vector.

Definition 14.1 (Vector-valued function). A function $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ takes a vector $\mathbf{x} \in \mathbb{R}^n$ and returns a vector $\mathbf{F}(\mathbf{x}) \in \mathbb{R}^m$. Writing out the m components,

$$\mathbf{F}(\mathbf{x}) = [F_1(\mathbf{x}), F_2(\mathbf{x}), \dots, F_m(\mathbf{x})],$$

where each **component function** $F_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a scalar function of the kind §10 introduced.

So a vector-valued function is nothing more than m scalar functions travelling together — the same packaging move as §12.1, applied to outputs instead of derivatives.

Example 14.1. $\mathbf{F}(x, y) = [x^2 - y^2, 2xy]$ is a function $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, with $F_1(x, y) = x^2 - y^2$ and $F_2(x, y) = 2xy$. (If those two components look familiar, they are the real and imaginary parts of z^2 for $z = x + iy$.)

14.2. The Jacobian

Each component F_i has n partial derivatives. With m components that is mn numbers — too many for a vector, and naturally arranged in a grid.

Definition 14.2 (Jacobian). Let $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ have all partial derivatives at $\mathbf{x}$. Its **Jacobian** is the $m \times n$ matrix

$$\mathbf{J}_{\mathbf{F}}(\mathbf{x}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \frac{\partial F_m}{\partial x_2} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}, \quad (\mathbf{J}_{\mathbf{F}})_{ij} = \frac{\partial F_i}{\partial x_j}.$$

It is also written $D\mathbf{F}$, and it is what “the derivative of $\mathbf{F}$ ” means.

Read the shape carefully: **one row per output, one column per input**. Row i collects all the partials of the single component F_i — so row i is the gradient of F_i . The Jacobian is a stack of gradients, one for each component function.

When $m = 1$ there is a single row, and the Jacobian is the gradient written across. The gradient was the Jacobian all along, in the case where the output was one number.

Example 14.2. For $\mathbf{F}(x, y) = [x^2 - y^2, 2xy]$, differentiating each component with respect to each variable gives the 2×2 matrix

$$\mathbf{J}_{\mathbf{F}}(x, y) = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}.$$

Note that this matrix is *not* symmetric — $-2y$ against $2y$. Jacobians in general have no symmetry to them.

Example 14.3. For $\mathbf{F}(x, y, z) = [xyz, x + y + z]$, a function $\mathbb{R}^3 \rightarrow \mathbb{R}^2$, the Jacobian is 2×3 :

$$\mathbf{J}_{\mathbf{F}}(x, y, z) = \begin{bmatrix} yz & xz & xy \\ 1 & 1 & 1 \end{bmatrix}.$$

Two outputs, three inputs, so two rows and three columns.

14.3. The gradient is such a function

Now return to $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Equation (2) recorded its gradient’s signature:

$$\nabla f : \mathbb{R}^n \longrightarrow \mathbb{R}^n.$$

That is exactly a vector-valued function of the kind just defined, with $m = n$. So it has a Jacobian — and that Jacobian is the second derivative we have been looking for.

15. The Hessian

15.1. The definition

Definition 15.1 (Hessian). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ have all second partial derivatives at $\mathbf{x}$. Its **Hessian** is the $n \times n$ matrix of those second partials,

$$\mathbf{H}_f(\mathbf{x}) = \mathbf{J}_{\nabla f}(\mathbf{x}), \quad (\mathbf{H}_f)_{ij} = \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

Written out for $n = 2$, with the variables called x and y :

$$\mathbf{H}_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x \partial x} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y \partial y} \end{bmatrix}.$$

The two entries on the diagonal are obtained by differentiating twice in the same variable. The two off the diagonal are the **mixed partials**: differentiate once in each variable, in the two possible orders.

Example 15.1. Take $f(x, y) = x^2y + 3y^2$, whose gradient was computed in Part 1, §7.2:

$$\nabla f(x, y) = [2xy, \quad x^2 + 6y].$$

Differentiate each component with respect to each variable:

$$\frac{\partial}{\partial x}(2xy) = 2y, \quad \frac{\partial}{\partial x}(x^2 + 6y) = 2x, \quad \frac{\partial}{\partial y}(2xy) = 2x, \quad \frac{\partial}{\partial y}(x^2 + 6y) = 6,$$

so

$$\mathbf{H}_f(x, y) = \begin{bmatrix} 2y & 2x \\ 2x & 6 \end{bmatrix}.$$

The two mixed partials both came out $2x$, even though they were computed by differentiating in opposite orders.

Example 15.2. Take $f(x, y, z) = x^2y + y^2z + z^2x$, so that

$$\nabla f = [2xy + z^2, \quad x^2 + 2yz, \quad y^2 + 2zx].$$

Differentiating each component in each of the three variables gives

$$\mathbf{H}_f(x, y, z) = \begin{bmatrix} 2y & 2x & 2z \\ 2x & 2z & 2y \\ 2z & 2y & 2x \end{bmatrix}.$$

Again every mixed pair agrees: the (1, 2) and (2, 1) entries are both $2x$, the (1, 3) and (3, 1) entries are both $2z$, and the (2, 3) and (3, 2) entries are both $2y$.

Twice is a coincidence. It is not.

15.2. Clairaut's theorem

Theorem 15.1 (Clairaut). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. If the mixed second partial derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ and $\frac{\partial^2 f}{\partial x_j \partial x_i}$ exist and are continuous near a point, then at that point they are equal:*

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}.$$

In words: *the order of differentiation does not matter*. Differentiating first with respect to x and then with respect to y gives the same result as doing it the other way round. This is not obvious — the two computations pass through entirely different intermediate expressions, as both examples above show — and it is why the mixed entries kept agreeing.

A common confusion (The continuity hypothesis is doing work). Clairaut's theorem is a theorem, not a definition, and it can fail. There exist functions whose two mixed partials both exist at a point and are *not* equal there; what such functions lack is the continuity of those mixed partials near the point. Every function you will meet in this course — polynomials, exponentials, logarithms, sines and cosines, and anything built from them by the rules of Part 1, §2 — satisfies the hypothesis comfortably, so the conclusion is always available. It is worth knowing that it rests on something.

15.3. The Hessian is symmetric

Clairaut's theorem is a statement about two numbers. Read as a statement about the matrix holding them, it says something clean.

The (i, j) entry of the Hessian is $\frac{\partial^2 f}{\partial x_i \partial x_j}$ and the (j, i) entry is $\frac{\partial^2 f}{\partial x_j \partial x_i}$. Clairaut says these are equal, for every pair i and j . So every entry of $\mathbf{H}_f$ equals its mirror image across the diagonal — which is precisely the definition of a symmetric matrix from the matrix unit, §6.5.

If the mixed second partial derivatives of f are continuous, then its Hessian is
symmetric:

$$\mathbf{H}_f = \mathbf{H}_f^\top.$$

Both worked examples above are symmetric, as they had to be. Contrast this with the Jacobian of §14.2, which had no such property: a general Jacobian need not even be square, let alone symmetric. Symmetry is not a feature of second derivatives in general — it is a consequence of the Hessian being the second derivative of a *scalar* function, so that its two indices play interchangeable roles.

Remark. This is worth registering as more than a curiosity. Symmetric matrices are easier to work with than general ones in almost every respect, and a great deal of what can be said about maxima, minima and the shape of a surface near a point depends on the Hessian being symmetric. An $n \times n$ symmetric matrix also carries only $n(n+1)/2$ independent numbers rather than n^2 , so storing one costs roughly half as much.

Looking ahead. The gradient told us which way f is going; the Hessian tells us how that direction is itself changing — the curvature of f . Where the gradient vanishes, it is the Hessian that decides whether the flat point is a minimum, a maximum, or neither. And an optimisation method that uses curvature rather than slope alone can take much better-informed steps than the one built on the gradient by itself.